



Linux User Management

This document provides a detailed explanation of Linux user management. User management is a critical skill in DevOps, Cloud Computing, and Cybersecurity, as it controls access, permissions, and system accountability.

1. Understanding Users in Linux

Linux is a multi-user operating system. Every action performed on a Linux system is executed by a user account. Proper user management ensures security, accountability, and controlled access to system resources.

There are three main types of users:

- Root user (administrator)
- System users (services and processes)
- Regular users (human users)

2. Important User Management Files

Linux stores user and group information in system files:

/etc/passwd – Stores user account information

/etc/shadow – Stores encrypted passwords

/etc/group – Stores group information

3. User Identification Commands

These commands display user identity and login information.

`whoami`

Displays the current logged-in user.

`id`

Shows user ID (UID), group ID (GID), and group memberships.

`who`

Displays users currently logged into the system.

4. Creating and Managing Users

`useradd username`

Creates a new user account.

`adduser username`

Interactive user creation command (recommended for beginners).

`passwd username`

Sets or changes a user's password.

`usermod -aG group username`

Adds a user to a supplementary group.

5. Deleting and Locking Users

```
userdel username
```

Deletes a user account.

```
userdel -r username
```

Deletes user account and home directory.

```
passwd -l username
```

Locks a user account.

```
passwd -u username
```

Unlocks a user account.

6. Group Management Commands

`groupadd groupname`

Creates a new group.

`groupdel groupname`

Deletes a group.

`groups username`

Displays groups a user belongs to.

7. Privilege Management (sudo)

`sudo command`

Executes a command with root privileges.

`visudo`

Safely edits the sudoers file.

Using sudo follows the principle of least privilege, which is critical in cloud and security environments.

8. User Management in DevOps, Cloud & Cybersecurity

DevOps: Controlled access to servers and CI/CD systems.

Cloud: Secure EC2 user access and SSH management.

Cybersecurity: Prevent unauthorized access and privilege escalation.